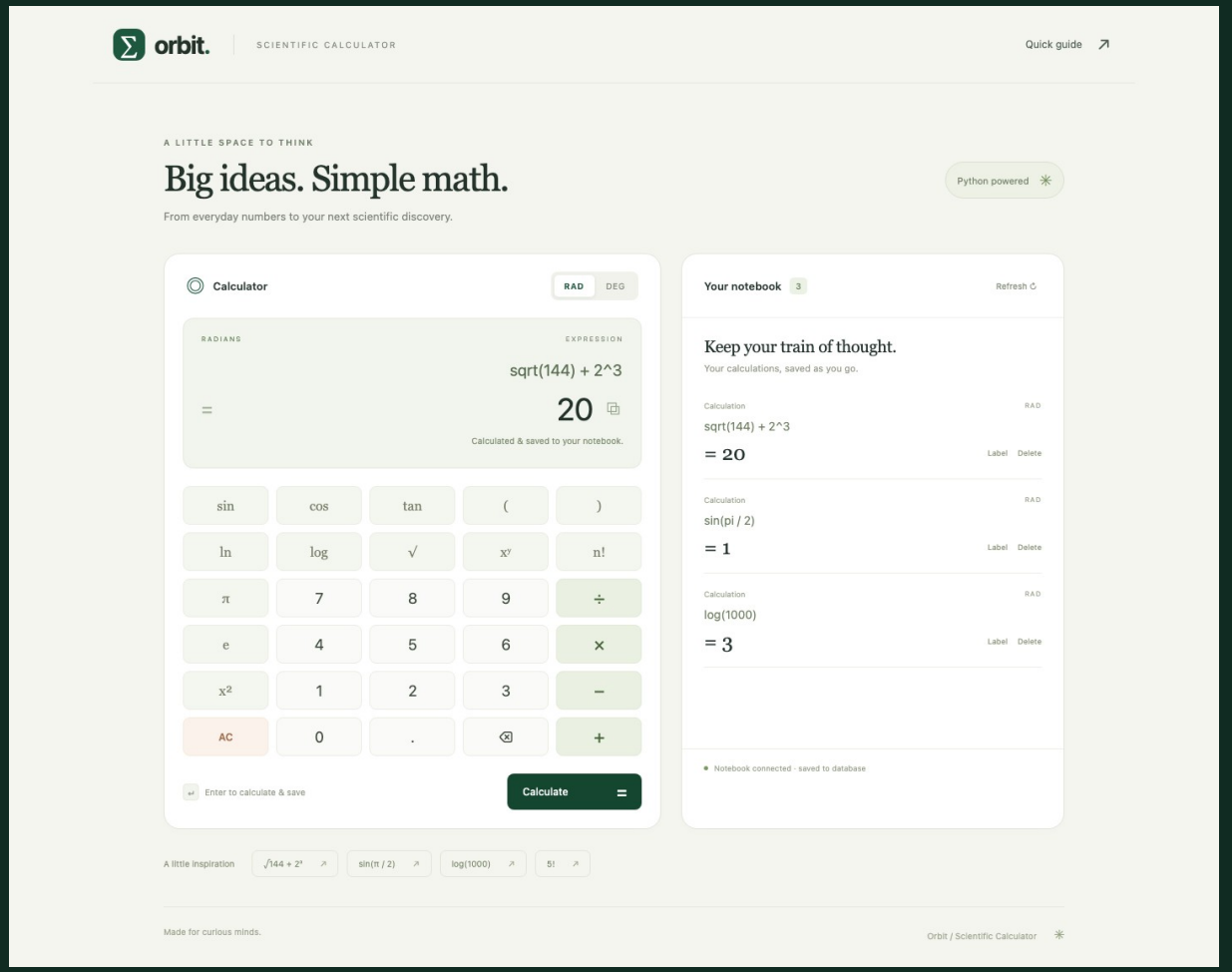
Orbit

Scientific calculator

DevOps final capstone

Python / FastAPI / PostgreSQL



Application architecture

Layer	Responsibility
HTML, CSS, JavaScript	Responsive calculator and notebook
Python FastAPI	Evaluate expressions and validate API requests
PostgreSQL + Alembic	Persist calculations and manage schema changes

REST API

GET, POST: /api/calculations
PUT, DELETE: /api/calculations/{id}

Health: /health Readiness: /ready Metrics: /metrics

Tests and code quality

69 tests passed

```
Orbit | All 69 local tests passing
Rendered transcript of actual output | source: pytest.txt

tests/test_api.py: 17 warnings
  /Users/yk/Desktop/coursework/DevOps assignment/DevOps-Capstone/.venv/lib/python3.14/site-packages/alembic/config.py:604:
DeprecationWarning: No path_separator found in configuration; falling back to legacy splitting on spaces, commas, and colons for
prepend_sys_path.  Consider adding path_separator=os to Alembic config.
    util.warn_deprecated(

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
====================== 69 passed, 18 warnings in 2.07s ======================
```

Isolated SQLite locally and disposable PostgreSQL in CI

Four frontend tests, a production build and Ruff checks also pass

Docker and local development

```
Orbit | docker-compose-status
Rendered transcript of actual output | source: docker-compose-status.txt

NAME               IMAGE                  COMMAND                  SERVICE    CREATED              STATUS
PORTS
orbit-backend-1    orbit-backend:local    "sh -c 'alembic upgr…"   backend    About a minute ago   Up About a minute (healthy)
127.0.0.1:8000->8000/tcp
orbit-frontend-1   orbit-frontend:local   "/docker-entrypoint.…"   frontend   About a minute ago   Up About a minute (healthy)
127.0.0.1:3000->8080/tcp
orbit-postgres-1   postgres:17-alpine     "docker-entrypoint.s…"   postgres   About a minute ago   Up About a minute (healthy)
5432/tcp
```

Three healthy services

Frontend, backend and PostgreSQL

Non-root application images

Backend UID 10001, frontend UID 101
Node build and Nginx runtime stages

GitHub Actions and GHCR

Version control

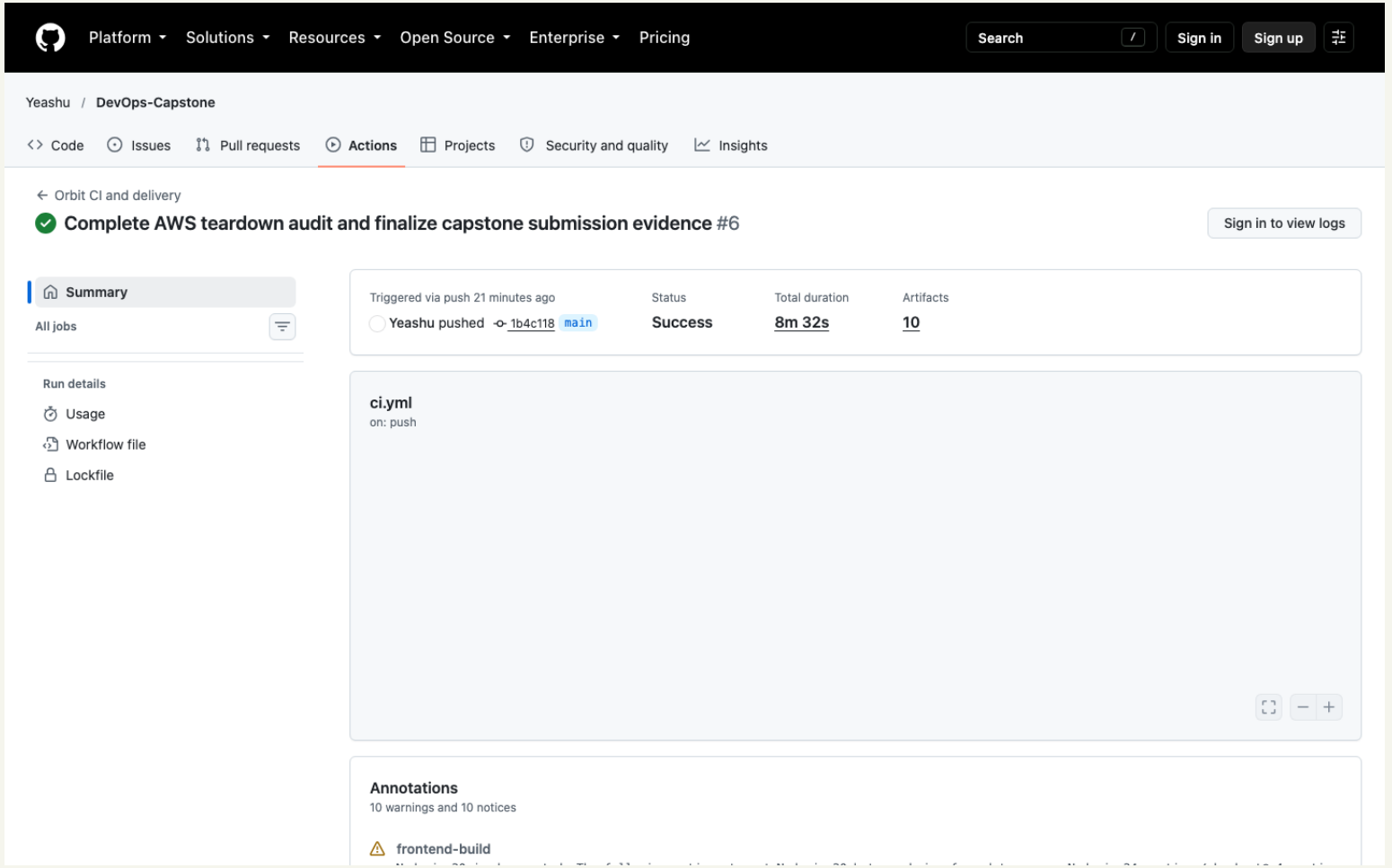
Public repository and meaningful commit history

Quality gates

Tests and both Trivy image scans precede publishing

Image publication

Exact scanned images use full commit SHA tags



CI result: green. Automatic EKS job: skipped.

Trivy security gate

0 HIGH or CRITICAL findings

Both remediated images pass the same strict gate

Orbit | Frontend scan: zero HIGH or CRITICAL findings

Rendered transcript of actual output | source: trivy-frontend.txt

Report Summary

Target	Type	Vulnerabilities
orbit-frontend:51064f19c877522e8ec4746193b1c9187953d0a8 (alpine 3.24.2)	alpine	0

Legend:
- '-': Not scanned
- '0': Clean (no security findings detected)

HIGH and CRITICAL findings fail CI, including unfixed vulnerabilities

Terraform and AWS EKS

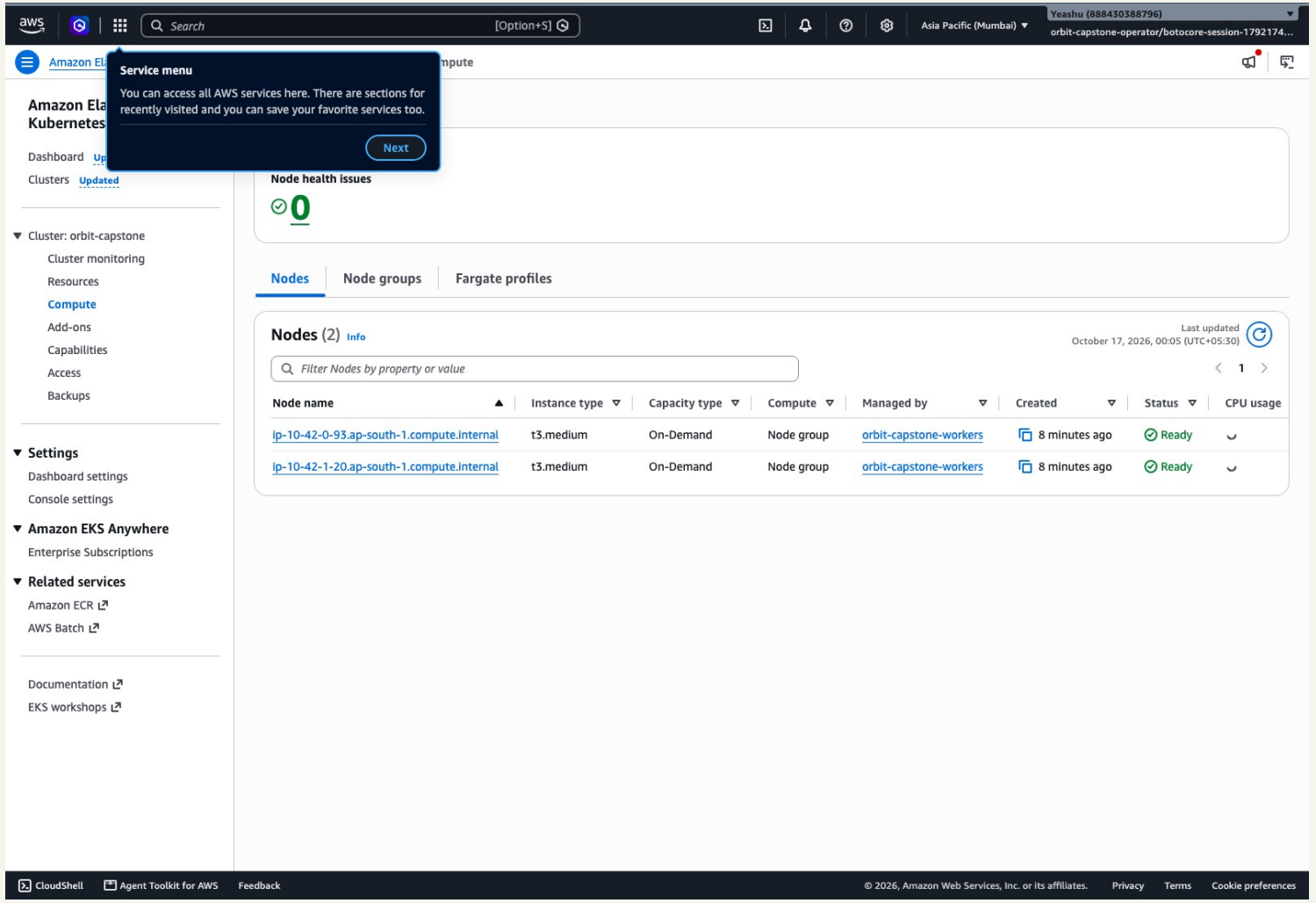
20

resources provisioned

Mumbai: ap-south-1

VPC and two public subnets

EKS with two Ready workers



Actual AWS capture before Terraform destroy

Kubernetes and Helm on EKS

```
Orbit | actual AWS EKS eks-pods
Rendered transcript of actual output | source: eks-pods.txt

NAME                              READY   STATUS      RESTARTS        AGE     IP            NODE
                                        NOMINATED NODE   READINESS GATES
orbit-backend-8555f778b5-4bpq5    1/1     Running     0               6m49s   10.42.0.86    ip-10-42-0-93.ap-
south-1.compute.internal   <none>           <none>
orbit-backend-8555f778b5-n9gj5    1/1     Running     0               6m49s   10.42.1.109   ip-10-42-1-20.ap-
south-1.compute.internal   <none>           <none>
orbit-frontend-77fc56bcc6-2nj74   1/1     Running     0               6m49s   10.42.1.164   ip-10-42-1-20.ap-
south-1.compute.internal   <none>           <none>
orbit-frontend-77fc56bcc6-l2j4z   1/1     Running     0               6m49s   10.42.0.191   ip-10-42-0-93.ap-
south-1.compute.internal   <none>           <none>
orbit-migrate-1-m7ksf             0/1     Completed   2 (6m36s ago)   6m49s   10.42.0.68    ip-10-42-0-93.ap-
south-1.compute.internal   <none>           <none>
orbit-postgres-0                  1/1     Running     0               6m49s   10.42.1.254   ip-10-42-1-20.ap-
south-1.compute.internal   <none>           <none>
```

2 backend replicas

2 frontend replicas

PostgreSQL storage

ClusterIP Services and Ingress routes for / and /api

Prometheus and Grafana

Application metrics

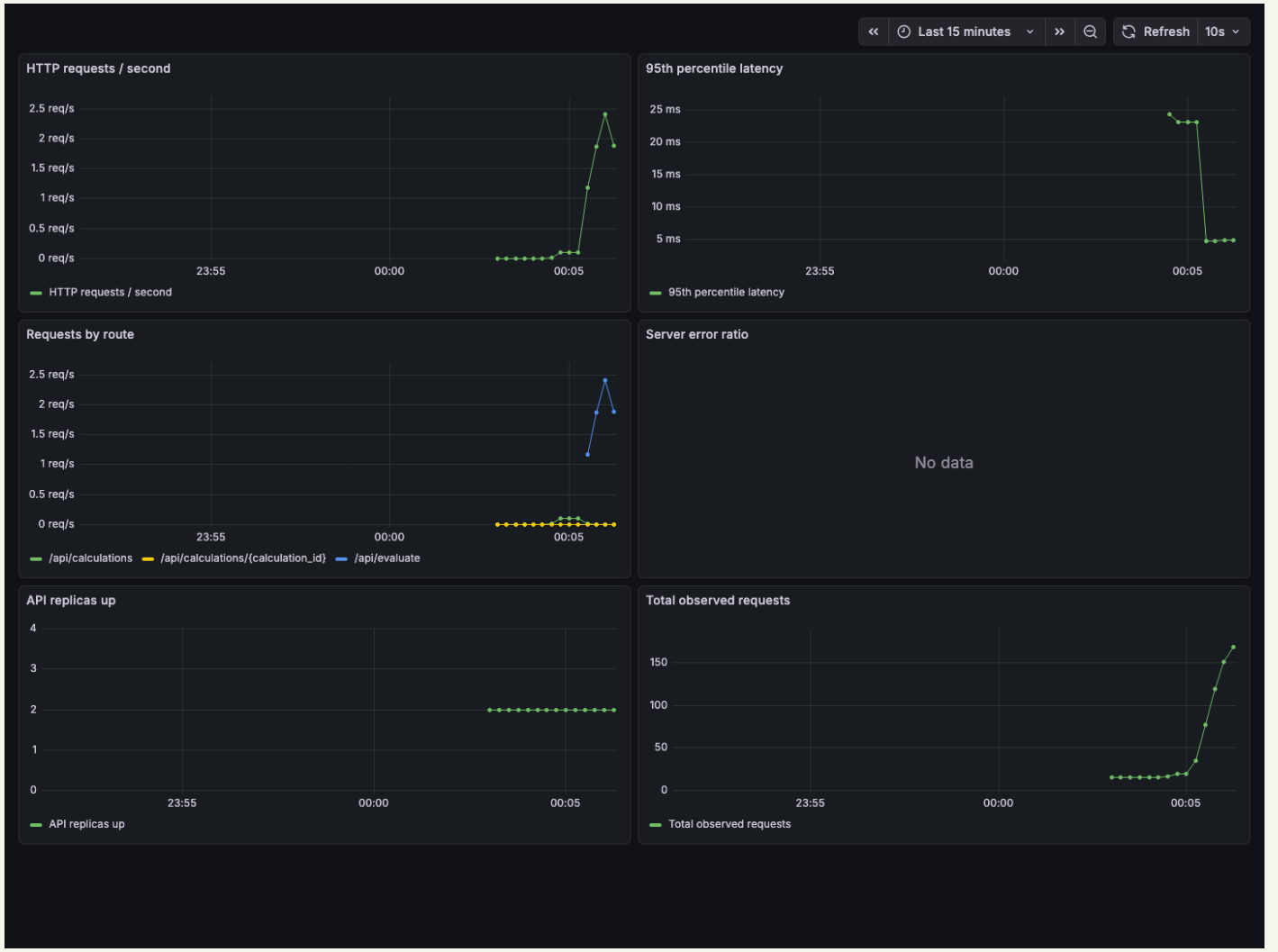
FastAPI exposes request and latency measurements

Prometheus

Both backend targets report UP

Grafana

Request and latency panels show data



Recorded deployment update

Successful pipeline

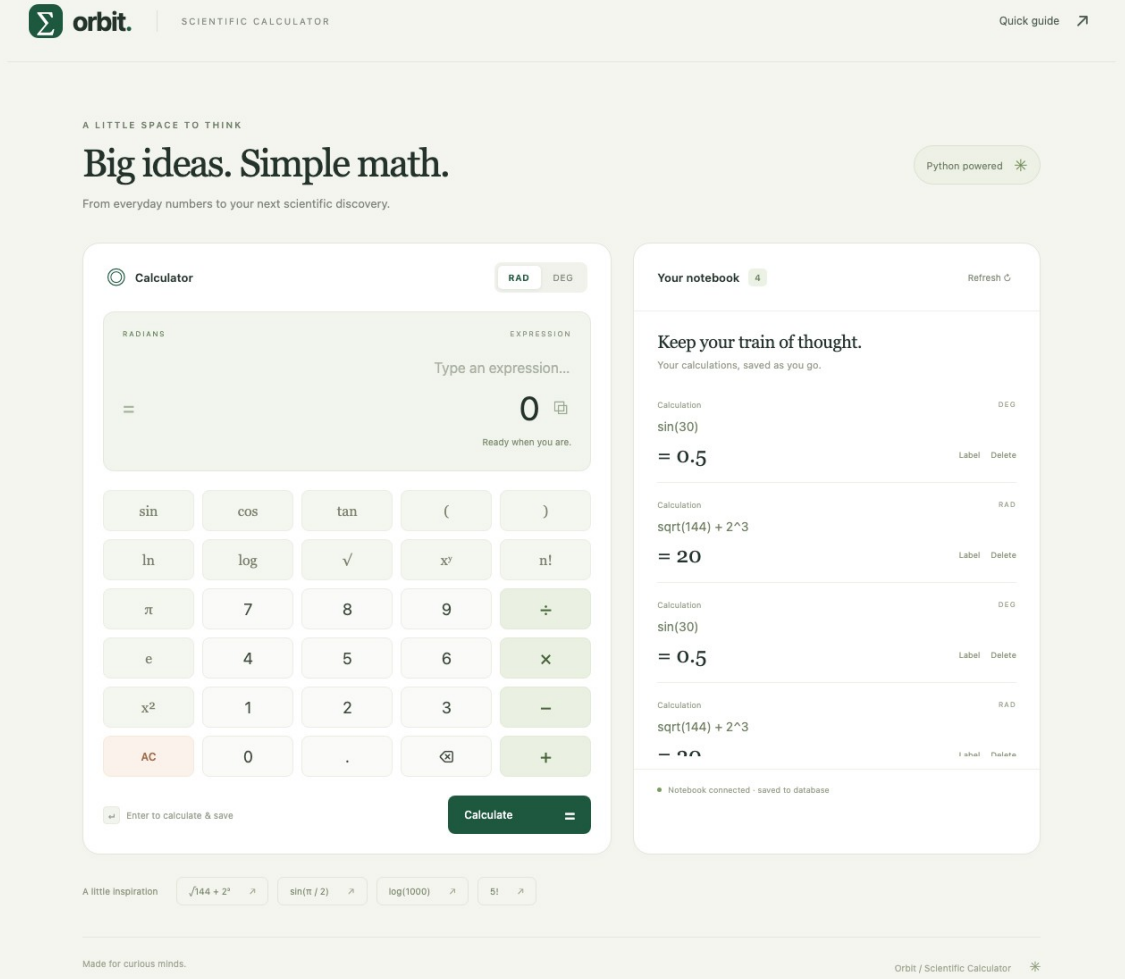
Run 38947611595 publishes the scanned SHA

Local Helm upgrade

EKS health reports commit 7f36caf

Persistent notebook

Calculations survive pod replacement



Local Helm delivery after green CI

Cleanup and submission

```
Orbit | actual Terraform destroy completion
Rendered transcript of actual output | source: terraform-destroy-summary.txt

Actual resumed Terraform destroy output excerpt. Plan and repetitive progress omitted.
Original lifecycle provisioned 20 resources; the first pass removed add-ons and AWS finished deleting the worker group after that
CLI process was interrupted.
The resumed destroy reconciled state and removed all remaining managed resources. See full logs and independent AWS cleanup audit.

aws_eks_cluster.main: Destroying... [id=orbit-capstone]
aws_route_table_association.public[1]: Destruction complete after 1s
aws_route_table_association.public[0]: Destruction complete after 1s
aws_route_table.public: Destroying... [id=rtb-0be1455cc55bd9266]
aws_route_table.public: Destruction complete after 0s
aws_internet_gateway.main: Destroying... [id=igw-0152cecee69e2508a]
aws_iam_role_policy_attachment.nodes["AmazonEKSWorkerNodePolicy"]: Destruction complete after 2s
aws_iam_role_policy_attachment.nodes["AmazonEC2ContainerRegistryReadOnly"]: Destruction complete after 2s
aws_iam_role_policy_attachment.nodes["AmazonEKS_CNI_Policy"]: Destruction complete after 2s
aws_iam_role.nodes: Destroying... [id=orbit-capstone-nodes]
aws_internet_gateway.main: Destruction complete after 1s
aws_iam_role.nodes: Destruction complete after 1s
aws_eks_cluster.main: Destruction complete after 3m7s
aws_iam_role_policy_attachment.cluster: Destroying... [id=orbit-capstone-cluster/arn:aws:iam::aws:policy/AmazonEKSClusterPolicy]
aws_subnet.public[0]: Destroying... [id=subnet-0e70e0cf94e8a5731]
aws_subnet.public[1]: Destroying... [id=subnet-01ae13c30f9d46046]
aws_subnet.public[1]: Destruction complete after 2s
aws_subnet.public[0]: Destruction complete after 2s
aws_vpc.main: Destroying... [id=vpc-0f918dcce080d2a5f]
aws_iam_role_policy_attachment.cluster: Destruction complete after 3s
aws_iam_role.cluster: Destroying... [id=orbit-capstone-cluster]
aws_vpc.main: Destruction complete after 1s
aws_iam_role.cluster: Destruction complete after 1s
Destroy complete! Resources: 14 destroyed.
```

Cloud teardown complete

Independent audit confirms no project EKS, VPC, active workers or volumes remain.

Source, 75 screenshots, the video and this deck remain in the repository.

github.com/Yeashu/DevOps-Capstone